

Thermo-sensitive polyethylene glycol-based hydrogel for controlled peptide decoy release: a novel therapy for bladder fibrosis in male mice with partial outlet obstruction

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ABSTRACT

A novel thermo-sensitive polyethylene glycol (PEG)-based hydrogel formulation combining KCF18—a peptide decoy that inhibits receptor binding of tumor necrosis factor- α , interleukin-1 β , and interleukin-6—was developed to enable controlled release of KCF18 for the treatment of bladder fibrosis resulting from partial bladder outlet obstruction (BOO). The hydrogel, composed of methyl PEG–poly(lactic-co-glycolic acid) (mPEG–PLGA, diblock copolymer) and PLGA–PEG–PLGA (triblock copolymer) at a 5:5 wt ratio, was mixed with KCF18 (1.5 mg/kg) to produce a 20 % KCF-hydrogel solution (100 μ L). Pharmacokinetic analysis revealed a peak plasma KCF18 concentration on Day 1, sustained through Day 2, with an estimated half-life of 3.5 days. In an adult male mouse model of BOO-induced bladder fibrosis, weekly subcutaneous administration of the KCF-hydrogel for seven weeks improved bladder function, reduced fibrosis, and downregulated transforming growth factor- β and fibrosis-related markers, including collagen I, vimentin, and α -smooth muscle actin. Therapeutic benefits were observed when treatment was initiated during the early or mid-disease stages (Day 7 or 14 post-BOO), but not in the late stage (Day 28). This thermo-sensitive PEG-based hydrogel enables controlled KCF18 release, offering a promising therapeutic strategy for BOO-induced bladder fibrosis when administered in the early or mid-disease stages.

1. Introduction

Benign prostatic enlargement is a key cause of bladder outlet obstruction (BOO) [1–4], typically managed with α -blockers and 5 α -reductase inhibitors [5]. However, patient adherence to these

treatments is often low [6], and without consistent intervention, BOO can lead to bladder remodeling, dysfunction, and fibrosis [7]. This progression causes urine retention, frequent urinary tract infections, and renal function impairment [8]. When standard treatments are ineffective, surgery might be an option, but it cannot fully reverse advanced

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bladder fibrosis [9–11]. Therefore, developing new treatments for BOO-induced bladder fibrosis to prevent irreversible bladder damage is crucial.

Increased intravesical pressure during urination due to outlet obstruction can cause ischemia-reperfusion injury in the bladder, leading to tissue inflammation [12,13]. Research shows that bladder inflammation from BOO is associated with elevated levels of inflammatory cytokines like tumor necrosis factor- α (TNF- α), interleukin-1 β (IL-1 β), and IL-6 [14,15], which are crucial in starting bladder remodeling, dysfunction, and fibrosis [15,16]. This involves activating transforming growth factor- β (TGF- β), a key regulator of fibrosis, in response to these cytokines [17]. Upon TGF- β activation, a complex series of fibrotic processes, including extracellular matrix (ECM) remodeling with excessive collagen production/deposition, epithelial-mesenchymal transition (EMT), and fibroblast to myofibroblast transition (FMT), initiated [18,19]. These insights highlight TNF- α , IL-1 β , and IL-6 as central to BOO-induced bladder fibrosis and potential therapeutic targets.

Peptide decoys, engineered to target and bind specific receptor ligands, exhibit high-affinity therapeutic effect and have shown effectiveness in treating disease like inflammatory diseases, cardiovascular diseases and breast cancer [20–23]. KCF18, a designed peptide decoy comprising 18 amino acids, incorporates N-terminal amino acids from TNF receptor 1 (TNFR1), middle amino acids from IL-1 receptor (IL-1R), and C-terminal amino acids from IL-6 receptor (IL-6R) [21]. KCF18 effectively blocks TNF- α , IL-1 β , and IL-6 receptor bindings, mimicking their receptors' binding sites [21]. Research has shown that KCF18 successfully treats lung and liver injuries in mice by blocking TNF- α , IL-1 β , and IL-6 receptor bindings [23], suggesting its potential effectiveness in treating BOO-induced bladder fibrosis.

Peptides' short in vivo half-life, due to rapid degradation and clearance, limits their use in chronic diseases like BOO-induced bladder fibrosis [24]. However, integrating them with polyethylene glycol (PEG)-based hydrogels can enhance their stability and lifespan [25]. These hydrogels can encapsulate peptides within a three-dimensional network, effectively restricting their diffusion and protecting them from rapid degradation, thereby improving in vivo retention and circulation time, while also enabling controlled release like the extracellular matrix [25,26]. For instance, a thermo-sensitive PEG-hydrogel system has been shown to extend monoclonal antibody release for up to 21 days [27]. Thermosensitive PEG-based hydrogels, with their injectability, biocompatibility, and physiological sol-gel transition behavior, are increasingly applied in therapy. Functional modifications further enhance their utility in disease models including cancer, fibrosis, and inflammation [28–32], supporting their use in the development of our KCF18-loaded hydrogel system.

In our study targeting BOO-induced bladder fibrosis, we tested a thermo-sensitive PEG-based hydrogel for the controlled release of KCF18 (KCF-hydrogel), hypothesizing continuous cytokine inhibition using KCF-hydrogel can alleviate bladder issues in a proven murine BOO model. We also examined the intervention's effects on related biological pathways. Additionally, this study also evaluated the impact of disease progression on the therapeutic efficacy of KCF18-hydrogel in this context.

2. Materials and methods

2.1. Animals

The goal of this study was to develop novel therapies for BOO-induced bladder fibrosis. As benign prostatic enlargement is the primary cause of BOO [1–4], adult male mice were selected for this investigation. Specifically, 8-week-old male BALB/c mice were obtained from the National Laboratory Animal Center, Taipei, Taiwan. Mice were housed under standard laboratory conditions with a 12-h light/12-h dark cycle and given free access to standard chow and water.

All animal procedures were reviewed and approved by the Institutional Animal Use and Care Committee of Taipei Medical University, Taipei, Taiwan (Approval No. LAC-2020-0434). Animal experiments were conducted in accordance with the ARRIVE guidelines, the U.K. Animals (Scientific Procedures) Act, 1986 and associated guidelines, the EU Directive 2010/63/EU for animal experiments, and the U.S. National Institutes of Health Guide for the Care and Use of Laboratory Animals (NIH Publication No. 8023, revised 1978).

2.2. Characteristics of KCF-hydrogel

2.2.1. Preparation of the KCF-hydrogel

KCF18, with a molecular weight of 2195.92 Da, is a peptide decoy derived from TNFR1, IL-1R, and IL-6R [21]. The “dock proteins protocol” (ZDOCK, BIOVIA Discovery Studio 3.5; BIOVIA, San Diego, CA, USA) was utilized for the design, and Mission Biotech (Taipei, Taiwan) synthesized KCF18 following established protocols [21].

The thermo-sensitive PEG-based hydrogels were prepared using methoxy PEG-poly(lactic-co-glycolic acid) (PLGA) (mPEG-PLGA, diblock copolymer, molecular weight 550–1400 Da) (Sigma-Aldrich, Burlington, MA, USA) and PLGA-PEG-PLGA (triblock copolymer, molecular weight 1250–1500–1250 Da) (Green Square, Taoyuan, Taiwan), combined in 5:5 or 4: 6 ratios, following established protocols [33]. Notably, prior rheological data showed that the gelling temperatures of the hydrogel formulation were 28 °C for the 5:5 ratio and 34 °C for the 4:6 ratio, with corresponding gel shrinkage of 10 % and 0 % at 37 °C [33].

The 20 % and 25 % KCF-hydrogel solutions (100 μ L) were prepared by mixing KCF18 (1.5 mg/kg) with the hydrogels at 4 °C overnight and stored at 4 °C. As a result, four distinct KCF-hydrogel formulations were obtained: 25 % KCF-hydrogel (5:5), 25 % KCF-hydrogel (4:6), 20 % KCF-hydrogel (5:5), and 20 % KCF-hydrogel (4:6). Notably, the 1.5 mg/kg dose of KCF18 was selected based on our previous findings showing that this dosage effectively inhibited the binding of TNF- α /TNFR, IL-1 β /IL-1R, and IL-6/IL-6R, thereby reducing sepsis-associated effects in endotoxin-treated mice [23].

2.2.2. In vitro release rate assay of KCF-hydrogel

The in vitro release assay was conducted based on a previously established method [34]. Briefly, the four KCF-hydrogel formulations (100 μ L each) were placed in centrifuge tubes, followed by the addition of 500 μ L of 10 mM phosphate-buffered saline (PBS, pH 7.4; Sigma-Aldrich). The tubes were incubated at 37 °C with continuous shaking at 100 rpm. Samples (500 μ L) were collected and replaced at 2, 4, 8 h, and days 1 and 2. KCF18 concentrations were measured using liquid chromatography–tandem mass spectrometry (Xevo TQ-S coupled with an ACQUITY UPLC pump system; Waters, Milford, USA) operated with MassLynx 4.1 software (Version 4.1, SCN 714; Waters) [23]. Mean KCF18 concentration at each time point was calculated from three independent batches of KCF-hydrogel.

The data were analyzed, and our results demonstrated that the 20 % KCF-hydrogel (5:5) formulation exhibited the best in vitro release efficacy (please refer to Fig. 1). This formulation was therefore selected for subsequent experiments to investigate its therapeutic effects on bladder fibrosis in BOO mice.

2.2.3. Structural and physicochemical characterization of KCF-hydrogel

The morphology of the KCF-hydrogel (20 %, 5:5) was examined using scanning electron microscopy (SEM) according to established procedures [35]. Briefly, hydrogel samples were stabilized by heating at 37 °C, frozen in liquid nitrogen, and lyophilized using a benchtop freeze-dryer (FD4.5-8P-D; Kingmech Scientific Co., LTD., New Taipei City, Taiwan). The freeze-dried KCF18-hydrogel specimens were sputter-coated with 2 nm of iridium and imaged with a scanning electron microscope (SU3500; Hitachi High Technologies Corp., Tokyo, Japan). A thermo-sensitive PEG-based hydrogel (20 %, 5:5) without the

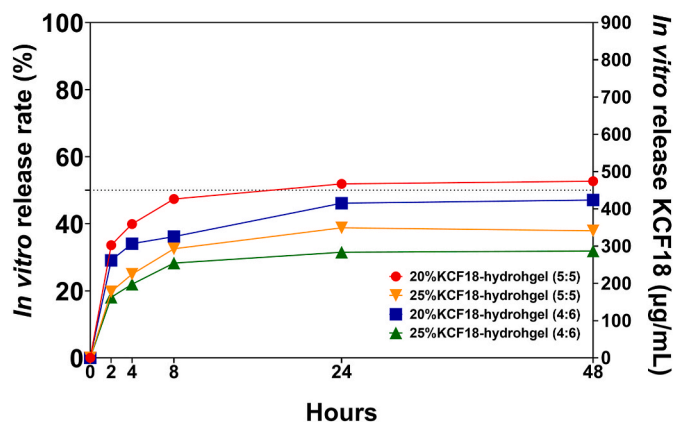


Fig. 1. In vitro release rates of four KCF-hydrogel formulations. These formulations—20 % (5:5), 25 % (5:5), 20 % (4:6), and 25 % (4:6)—were prepared by combining methyl polyethylene glycol (mPEG)-poly(lactic-co-glycolic acid) (PLGA) (mPEG-PLGA, diblock copolymer) and PLGA-PEG-PLGA (triblock copolymer) in 5:5 and 4:6 ratios to form thermos-sensitive PEG-based hydrogels. Each hydrogel was then conjugated with KCF18 (1.5 mg/kg). Mean KCF18 concentration at each time point, calculated from three independent batches of KCF-hydrogel, is presented. The left Y-axis represents the cumulative release rate (%) of KCF18, while the right Y-axis shows the cumulative concentration of KCF18 (µg/mL).

KCF18 peptide decoy was processed in parallel and used as the control.

Physicochemical characterization of the KCF18-hydrogel (20 %, 5:5) was performed following established protocols [33], including phase diagram analysis, shrinkage behavior, and rheological measurements. A peptide-free hydrogel of the same composition (20 %, 5:5) was used as the control in all assessments. For phase diagram analysis, the transition between the liquid (sol) and gel states was evaluated as a function of polymer concentration and temperature (15–45 °C). Hydrogel samples were gradually heated from 15 to 45 °C at a controlled rate of 1 °C/min. The sol–gel transition temperature was determined using the tube inversion method, defined as the lowest temperature at which no flow was observed upon inversion. For shrinkage behavior analysis, hydrogel samples (1 mL each) were incubated at 37 °C for 6 h, and shrinkage ratios were calculated based on the volume of water expelled. Rheological measurements were performed on an MCR 302 rheometer equipped with a CP25-1 spindle (Anton Paar, Graz, Austria). Measurements were conducted over a temperature range of 15–45 °C, with a heating rate of 1 °C/min, frequency of 10 rad/s, and strain of 0.5 %.

2.2.4. Pharmacokinetic analysis

A set of mice ($n = 33$) was allocated for the pharmacokinetic analysis of the KCF-hydrogel (20 %, 5:5). Each mouse received a single subcutaneous injection of the KCF-hydrogel (100 µL) at the back of the neck. Blood samples were collected via cardiac puncture before administration (time 0) and at 30 min, 8 and 16 h, and on Days 1, 2, 3, 4, 5, 6, and 7 post-administrations ($n = 3$ per time point). At each sampling point, approximately 750–900 µL of blood sample was collected per mouse under isoflurane anesthesia. After collection, the samples were immediately centrifuged at 3000 rpm for 10 min using the Eppendorf 5415R Refrigerated Centrifuge (Sigma-Aldrich) at room temperature. The resulting plasma was stored at −80 °C for subsequent analysis. Plasma concentrations of KCF18 were measured using liquid chromatography–tandem mass spectrometry (Xevo TQ-S with ACQUITY UPLC pump system; Waters), as described in Section 2.2.2 [23]. Pharmacokinetic parameters were calculated using PKSolver, an add-in program for Microsoft Excel, based on a non-compartmental model.

2.2.5. Biodistribution of KCF18 using ex vivo bioluminescence imaging

A set of mice ($n = 15$) was utilized to assess the biodistribution of KCF18 following a single subcutaneous injection of the KCF-hydrogel

(100 µL; 20 %, 5:5). For imaging purposes, KCF18 peptide was conjugated with rhodamine B (Abcam, Cambridge, UK). After the injection of the rhodamine B-conjugated KCF-hydrogel (100 µL) at the back of the neck, mice were euthanized by decapitation before administration (0 h) and on Days 1, 3, 5, and 7 following the KCF-hydrogel injection ($n = 3$ per time point), and the heart, lungs, liver, kidney, spleen, and bladder were subsequently harvested. Biodistribution of KCF18 was evaluated through the fluorescence signals of rhodamine B using an in vivo imaging system (IVIS Lumina XRMS; PerkinElmer, Waltham, MA, USA) for bioluminescence imaging [23]. Subsequently, the images were analyzed, and signal intensities were measured using Living Image software (PerkinElmer, Waltham, MA, USA). Moreover, to confirm signal specificity, a separate set of mice ($n = 15$) received rhodamine B-only hydrogel, and the biodistribution of rhodamine B was assessed using the same method described above.

2.3. Bladder fibrosis progression in BOO mice

2.3.1. The surgical procedures of BOO

The protocols for the BOO surgical procedures were adapted from a previous report [36]. With anesthesia (zoletil/xylazine, 40/10 mg/kg, intraperitoneal injection), mice received a 0.5-cm midline vertical incision, extending from the penoscrotal junction to the mid-scrotum, to access the bulbous urethra. Following isolation from the cavernous bodies, the urethral surface was exposed, and a sterile metal bar with a 0.3-mm diameter was positioned. A 4-0 vicryl suture (J504G; Ethicon Inc., Bridgewater, NJ, USA) was used to create a tie around both the urethra and the metal bar. Subsequently, the metal bar was removed, leaving the urethra partially obstructed. The muscle layer was re-approximated, and the skin closed using a 3-0 nylon suture (663G; Ethicon Inc.). The incision wound was cleansed with a 10 % povidone iodine solution (10 % w/v; Betadine®. Win-Medicare Pvt Ltd., New Delhi, India), and immediate postoperative analgesia was administered with local injection of bupivacaine (0.5 %, 100 µL; Pfizer, New York, NY, USA). The mice were placed in cages and maintained with a heating lamp. As a control for the surgical procedures, a separate group of anesthetized mice underwent a sham operation, involving a penoscrotal incision and urethra isolation without the placement of a urethral tie.

2.3.2. BOO-induced bladder function deterioration measured using mice voiding behavior analysis

To confirm the development and progression of bladder fibrosis in this BOO murine model, a set of 15 mice received BOO surgery and 5 mice received sham operation. On days 7, 14, and 28 post-BOO, the bladder function of the BOO mice (5 mice per time point) was evaluated using voiding behavior analysis (designated as the BOO-7D, BOO-14D and BOO-28D groups, respectively), as per established methods [37]. Voiding behavior analysis was also conducted in sham-operated mice (Sham group). Briefly, on the day of evaluation, each mouse was individually housed in a cage furnished with standard rodent chow and lined with a 0.35-mm filter paper (26.5 × 21 cm; No. GFP001; GenePure, Taichung, Taiwan) for a duration of 4 h in a dark room. Urine spots on the filter papers were captured using a documentation system (UVP UVsolo touch darkroom ultraviolet imaging cabinet; Analytik, Jena, Germany). Image capture settings were adjusted using the UVP Vision-Works touch image acquisition software (Jena). Images were obtained using the ninhydrin test (1 % ninhydrin in 75 % ethanol solution; Sigma-Aldrich) and 365-nm UV illumination. The exposure settings were optimized to enhance the signal-to-noise ratio, following which the counts of urine spots (urinary frequency) and areas of urine spots (bladder capacity) were recorded.

2.3.3. BOO-induced bladder wall fibrosis measured using Masson's trichrome staining

After voiding behavior analysis, all mice were euthanized via decapitation, and their bladders were harvested. One half was snap

frozen and stored at -80°C for later use. The other half was fixed in 4 % paraformaldehyde, embedded in paraffin (both from Sigma-Aldrich), sectioned, and mounted on slides. Subsequently, Masson's trichrome stain (Trichrome Stain Kit; Sigma-Aldrich) was employed to visualize the collagen (blue staining) in the bladder tissues, allowing for the evaluation of fibrotic status, as previously detailed [19]. Six random fields from each sample were chosen under a light microscope (MoticEasyScan Pro 6; Motic Asia, Hong Kong) to facilitate quantification of the density of Masson's trichrome staining. The staining density was analyzed using ImageJ software (version 1.52e.), and the average staining density value of the six fields in each section was recorded.

2.3.4. Expression of cytokines measured using enzyme-linked immunosorbent assay (ELISA)

The expression of cytokines in bladder tissues was assessed using ELISA. Briefly, snap frozen bladder tissues were thawed and processed, as per a previous report [23]. The concentrations of TNF- α , IL-1 β , and IL-6 were then analyzed using ELISA (ELISA kits for TNF- α , IL-1 β , and IL-6; all from Sigma-Aldrich), as per the manufacturer's protocols.

2.3.5. Expression of fibrosis markers measured using immunoblotting assay

Snap-frozen bladder tissues were thawed, homogenized, micro-waved, and centrifuged to collect the supernatants for analysis, following a previous report [23]. The protein concentration of each sample was determined using a bicinchoninic acid protein assay kit (Pierce Chemical Co., Rockford, IL, USA). An equal amount of protein (20 μg) from each sample was loaded to sodium dodecyl sulphate-polyacrylamide gels (12 %; Sigma-Aldrich) and separated by electrophoresis and transferred to nitrocellulose membranes (Bio-Rad Laboratories, Hercules, CA, USA). Subsequently, the membranes were incubated overnight with primary antibodies against TGF- β (dilution 1:1000, ab215715; Abcam, Cambridge, UK), the ECM remodeling marker collagen I (dilution 1:200, GTX26308; GeneTex, Hsinchu, Taiwan), the FMT marker α -smooth muscle actin (α -SMA) (dilution 1:400, A2547; Sigma-Aldrich), the EMT marker vimentin (dilution 1:500, AX10005; Abcepta, San Diego, CA, USA) [18,19], or the internal standard actin (dilution 1:5000, A2066; Sigma-Aldrich). Bound antibodies were detected by chemiluminescence using the ECL Plus kit (Amersham BioSciences, Buckinghamshire, UK). Images were captured with a multispectral imaging system (UVP BioSpectrum 815) and analyzed using VisionWorks LS software (Analytik Jena).

2.4. Therapeutic effects of KCF-hydrogel against bladder fibrosis in BOO mice

To investigate the therapeutic effects of KCF-hydrogel, a treatment regimen involving weekly subcutaneous injections of KCF-hydrogel (100 μL ; 20 %, 5:5), containing 1.5 mg/kg KCF18, at the back of the neck for 7 consecutive weeks was designed.

The KCF18 dosage used in this study, as previously mentioned, was initially determined based on our earlier findings that 1.5 mg/kg KCF18 effectively inhibits the binding of TNF- α /TNFR, IL-1 β /IL-1R, and IL-6/IL-6R [23]. Additionally, we conducted a preliminary study to evaluate the therapeutic effects of KCF-hydrogel containing KCF18 at a lower dose (0.6 mg/kg). The results showed that the 0.6 mg/kg dose did not produce significant improvements in bladder dysfunction or fibrosis (Supplemental Fig. 1C). Based on these findings, the 1.5 mg/kg dose was selected for further investigation in this study.

In addition, our data showed that KCF-hydrogel facilitated the sustained release of KCF18 for up to 7 days (please refer to Fig. 3). Furthermore, we observed significant bladder dysfunction and bladder wall fibrosis by Day 7 following BOO induction (early stage), with worsening symptoms by Days 14 and 28 (mid- and late disease stages) (please refer to Figs. 5 and 6). Based on these observations, the therapeutic regimen was initiated on Day 7 (early stage), Day 14 (mid stage), or Day 28 (late stage) following BOO to assess the therapeutic effects of

KCF-hydrogel at different stages of BOO-induced bladder fibrosis.

2.4.1. Therapeutic effects of KCF-hydrogel, initiated in the early stage of bladder fibrosis, against BOO-induced bladder fibrosis in mice

2.4.1.1. Effects of KCF-hydrogel on inhibiting cytokine-receptor binding in bladders of BOO mice.

To confirm the effects of KCF-hydrogel on inhibiting cytokine-receptor binding in bladders of BOO mice, two groups of mice were used, including BOO and BOO-7D + KCF-H groups ($n = 4$ in each group). Mice in the BOO-7D + KCF-H group received weekly subcutaneous injections of the KCF-hydrogel (100 μL ; 20 %, 5:5) administered at the back of the neck for 7 consecutive weeks, starting on Day 7 post-BOO, while mice in the BOO received normal saline (100 μL) in the same manner. On Day 56 post-BOO, all mice were euthanized via decapitation and the bladder was harvested. The BOO-induced binding of TNF- α to TNFR1, IL-1 β to IL-1R, and IL-6 to IL-6R in bladder tissues were assessed via immunofluorescence staining using the proximity ligation assay (PLA) [23].

In brief, the DuoLink Mouse Rabbit in situ PLA kit (Sigma-Aldrich) was utilized [23] and performed in accordance with the manufacturer's instructions. Bladder tissue sections were first blocked with Duolink® blocking solution and then incubated with primary antibodies for TNF- α (1:200 dilution, ab183218; Abcam), TNFR1 (1:200 dilution, ab60031; Abcam), IL-1 β (1:200 dilution, ab283818; Abcam), IL-1R (1:200 dilution, 50807-R075; Sino Biological, Beijing, China), IL-6 (1:200 dilution, 10395-MM19; Sino Biological) and IL-6R (1:200 dilution, ab300581; Abcam), respectively. This was followed by incubation with the oligonucleotide-labeled secondary antibody (PLA probe, Sigma-Aldrich). Subsequently, the tissue sections were stained with DAPI (1:200 dilution, Sigma-Aldrich) to visualize the nuclei. Laser scanning microscopy (Leica TCS SP8 CARS; Leica Microsystem, Wetzlar, Germany) equipped with a 25x water immersion objective (Fluotar VISIR 25x/0.95 WATER; Leica microsystems) was used to scan all sections. Image J was employed to analyze the resultant images.

2.4.1.2. Therapeutic effects of KCF-hydrogel against BOO-induced bladder fibrosis.

A set of 60 mice were randomly allocated to the following 6 groups, including the Sham, Sham + H, Sham + KCF-H, BOO, BOO-7D + H, and BOO-7D + KCF-H groups ($n = 10$ per group). Mice were randomly assigned to undergo either a sham or BOO surgery on the first day of the experiment, following the above-described procedure in Section 2.3.1. Beginning on Day 7 post-surgery, mice in the Sham + KCF-H and BOO-7D + KCF-H groups received a subcutaneous injection of KCF-hydrogel (100 μL ; 20 %, 5:5) once per week for 7 consecutive weeks, as above described. To serve as controls for the injection and vehicle, mice in the Sham and BOO groups received subcutaneous injections of normal saline (100 μL), while mice in the Sham + H and BOO-7D + H groups received subcutaneous injections of a corresponding thermo-sensitive PEG-based hydrogel (100 μL ; 20 %, 5:5), also once per week for 7 consecutive weeks. All mice were closely monitored and sacrificed via decapitation on the 56th day post-surgery.

2.4.1.2.1. Bladder function and fibrosis assessments.

The bladder function deterioration assessment was conducted in 5 mice from each group via analysis of mice voiding behavior. On Day 56 post-surgery, following the protocol outlined in Section 2.3.2, the counts and areas of urine spots were recorded and subsequently, voiding behavior was analyzed and compared.

After the voiding behavior measurement was completed, the same set of mice were euthanized via decapitation, and their bladders were harvested. The fibrosis status of the bladders was assessed using Masson's trichrome staining ($n = 5$ in each group), following the protocol outlined in Section 2.3.3.

2.4.1.2.2. Expression evaluation of cytokines.

The remaining 5 mice from each group were sacrificed via decapitation on the 56th day post-surgery. The bladder was harvested, snap frozen and stored at -80°C .

The expression levels of cytokines, including TNF- α , IL-1 β , IL-6 and TGF- β , in bladder tissues were analyzed using ELISA, following the procedure outlined in Section 2.3.4.

2.4.1.2.3. Expression evaluation of fibrosis markers. The expression levels of fibrosis markers, including TGF- β , collagen I, vimentin, E-cadherin and α -SMA, in snap frozen bladder tissues were analyzed using immunoblotting assay, following the procedure outlined in Section 2.3.5.

Additionally, the expression levels of fibrosis markers were analyzed using immunohistochemistry staining assay, following previously established method [23]. Paraffin-embedded bladder tissues were sectioned and then incubated with primary antibodies against TGF- β (dilution 1:500, ab215715; Abcam), collagen I (dilution 1:100, GTX26308; GeneTex), α -SMA (dilution 1:200, A2547; Sigma-Aldrich), or vimentin (dilution 1:250, AX10005; Abcepta). Six random fields per sample were examined (MoticEasyScan Pro 6; Motic Asia), and the densities were analyzed (ImageJ software) with the average staining density recorded.

2.4.2. Therapeutic effects of KCF-hydrogel, initiated in the mid- and late stages of bladder fibrosis, against BOO-induced bladder fibrosis in mice

For mid- and late stage effects, two additional sets of 10 mice each underwent BOO surgery, with KCF18-hydrogel (100 μ L; 20 %, 5:5) treatment initiated on Days 14 and 28 respectively, and designated as the BOO-14D + KCF-H and BOO-28D + KCF-H groups. The mice were monitored and euthanized after 7 weeks of treatment. Evaluations of therapeutic effects were performed on Days 63 and 77 respectively, through bladder function deterioration assessed by voiding behavior analysis (as described in Section 2.3.2), bladder wall fibrosis assessed by Masson's trichrome stain (as described in Section 2.3.3), cytokine expression analyzed by ELISA (as described in Section 2.3.4), and fibrosis marker expression measured using immunohistochemistry staining assay (as described in Section 2.4.2.3).

2.5. Statistical analysis

A one-way analysis of variance (ANOVA) was employed to evaluate group differences, followed by post-hoc analysis using the Student-Newman-Keuls test. The data were presented as means $\pm$ standard deviations, with a significance level set at 0.05. Data analysis was conducted using the Prism package in GraphPad (GraphPad Software, Inc., La Jolla, CA, USA).

3. Results

3.1. Characteristics of the KCF-hydrogel

Fig. 1 shows the in vitro release profile of the four KCF-hydrogel formulations. All formulations exhibited a similar pattern, characterized by a rapid increase in release within the first 8 h, followed by a slower increase through Days 1–2. Notably, the 20 % KCF-hydrogel (5:5) formulation demonstrated the highest in vitro release efficacy, with approximately 55 % released by Day 2. In comparison, the other three formulations released approximately 30–35 % by 8 h and around 35–45 % by Day 2. Based on these results, the 20 % KCF-hydrogel (5:5) formulation was selected for subsequent experiments.

Fig. 2 presents the structural characteristics of the KCF-hydrogel (20 %, 5:5) and its corresponding control hydrogel, as observed by scanning electron microscopy. Fig. 3 illustrates the physicochemical properties of both hydrogels, including (A) phase diagram analysis, (B) shrinkage behavior, (C) representative photographs of both hydrogels using the tube inversion method, and (D) rheological measurements. The results indicate that incorporation of the KCF18 peptide did not significantly affect these parameters, confirming that the fundamental physicochemical characteristics of the hydrogel were preserved.

Fig. 4 shows the plasma concentrations of KCF18, which peaked at approximately 24 h (Day 1), plateaued until 48 h (Day 2), and then gradually declined, with KCF18 remaining detectable up to 144 h (Day 6). Pharmacokinetic analysis revealed a maximum plasma concentration (C_{max}) of 20.53 ng/mL, with the time to reach maximum concentration (T_{max}) occurring at 48 h. The estimated in vivo half-life was approximately 84 h (3.5 days), the area under the concentration–time curve (AUC) was 1805.55 ng h/mL, and the mean residence time (MRT) was 58 h.

Fig. 5 shows KCF18 biodistribution in mice, with fluorescence signals indicating its presence in organs. Increased fluorescence was observed in the liver, kidney, and bladder, particularly the kidney, while the heart, lungs, and spleen had low signals. Quantitative analysis showed significant fluorescence in the kidney on Days 1, 3, and 5 ($p = 0.022$, $p = 0.006$ and $p = 0.002$, respectively), in the liver on Day 1 ($p = 0.006$), and in the bladder on Day 3 ($p = 0.037$). No significant differences were found in the heart, lungs or spleen. Moreover, IVIS imaging of mice treated with rhodamine B-only hydrogel showed minimal and transient fluorescence (please refer to Supplemental Fig. 1A). These data confirm that the fluorescent signals were emitted by rhodamine B-KCF, but not rhodamine B hydrolyzed from the conjugate.

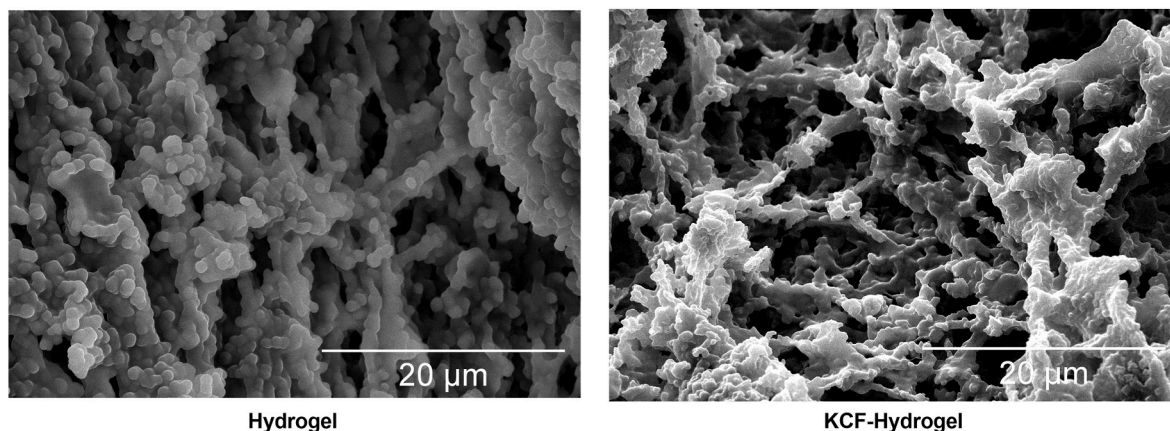


Fig. 2. Representative scanning electron microscopy images of the hydrogel and KCF-hydrogel. The hydrogel is a thermos-sensitive polyethylene glycol (PEG)-based formulation (20 %, 5:5). The KCF-hydrogel consists of the same PEG-based hydrogel (20 %, 5:5) conjugated with KCF18 (1.5 mg/kg).

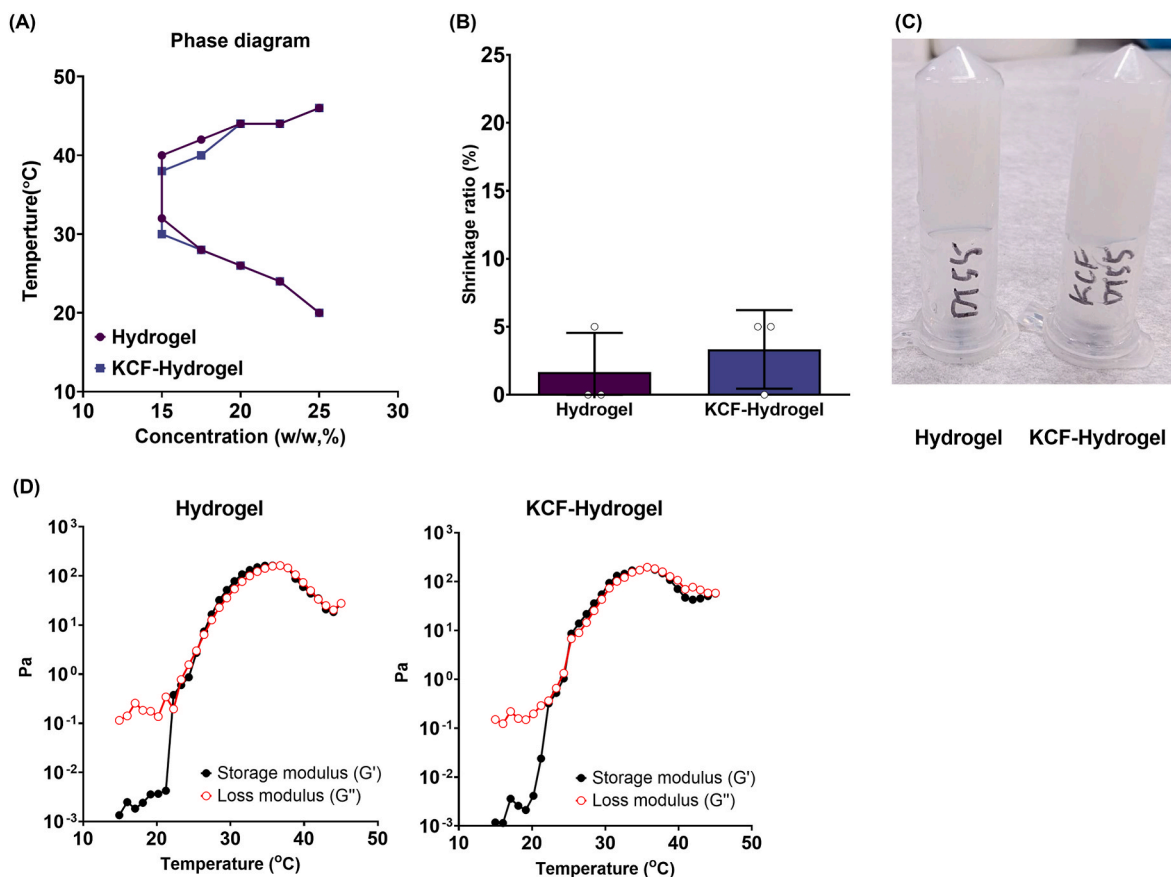


Fig. 3. Physicochemical characterization of the hydrogel and KCF-hydrogel. The hydrogel is a thermo-sensitive polyethylene glycol (PEG)-based formulation (20 %, 5:5), while the KCF-hydrogel consists of the same PEG-based hydrogel (20 %, 5:5) conjugated with KCF18 (1.5 mg/kg). (A) Phase diagrams, representing the sol-gel transition temperatures of the hydrogel and KCF-hydrogel, showing the transition between liquid (sol) and gel states as a function of polymer concentration and temperature (15–45 °C). (B) Shrinkage ratios of the hydrogel and KCF-hydrogel after 6 h of incubation at 37 °C. (C) Representative photographs of the hydrogel and KCF-hydrogel in their gelled state at 37 °C, using the tube inversion method. (D) Temperature-dependent rheological profiles showing the storage modulus (G') and loss modulus (G'') for the hydrogel and KCF-hydrogel, demonstrating comparable rheological behavior.

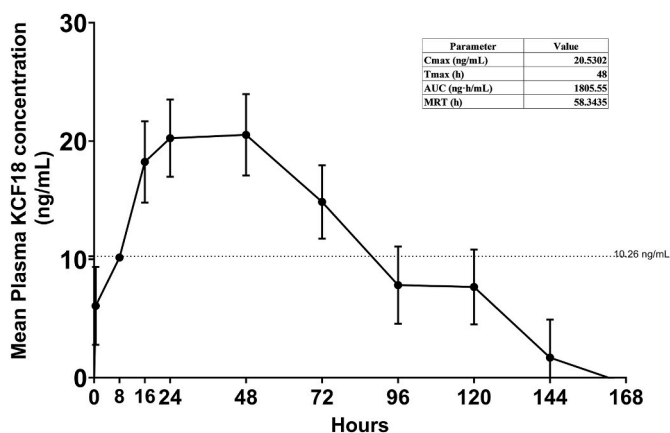


Fig. 4. Pharmacokinetic analysis of KCF-hydrogel. A total of 33 adult male mice received a single subcutaneous injection of KCF-hydrogel (100 μ L), containing 1.5 mg/kg KCF18 in a thermos-sensitive polyethylene glycol-based hydrogel (20 %, 5:5). Blood samples were collected via cardiac puncture at baseline (time 0) and at 30 min, 8 h, 16 h, and on Days 1–7 post-injection ($n = 3$ per time point). Plasma KCF18 concentrations were measured using liquid chromatography–tandem mass spectrometry and expressed as mean $\pm$ standard deviation. Pharmacokinetic profile of KCF18 after subcutaneous KCF-hydrogel injection was then calculated. C_{max}: the maximum plasma concentration. T_{max}: the time to reach maximum concentration. AUC: the area under the concentration–time curve. MRT: the mean residence time.

3.2. Bladder fibrosis progression in BOO mice

3.2.1. Progression of bladder dysfunction and fibrosis in BOO mice

Fig. 6(A) shows voiding behavior analysis in male mice at Days 7, 14, and 28 post-BOO. Compared to the Sham group, urine spot count was higher and spot area smaller in all BOO groups ($p < 0.001$ for all). The BOO-14D and BOO-28D groups had more spots than the BOO-7D group ($p = 0.011$ and $p < 0.001$, respectively), and the BOO-28D group had smaller spot area compared to BOO-7D ($p = 0.004$). The BOO-28D group had more spots and smaller spot areas than the BOO-14D group ($p = 0.028$ and $p < 0.001$, respectively). These findings show BOO-induced bladder dysfunction worsens over time.

Fig. 6(B) shows Masson's trichrome staining for collagen deposition. Staining was similar in Sham and BOO-7D groups but higher in BOO-14D and BOO-28D groups ($p < 0.001$ and $p = 0.001$, respectively). No differences were found between BOO-14D and BOO-28D groups. This indicates progressive bladder wall fibrosis.

3.2.2. Progression of cytokine and fibrosis marker expression in bladders of BOO mice

Fig. 7(A) shows cytokine expression by ELISA. IL-1 β levels were significantly higher in all BOO groups compared to Sham (all $p < 0.001$). IL-1 β levels increased from BOO-7D to BOO-28D ($p < 0.001$), with the BOO-28D group showing higher levels than BOO-14D ($p = 0.004$). IL-6 followed a similar trend, and TNF- α was significantly higher in all BOO groups compared to Sham ($p = 0.035$, $p = 0.020$ and $p = 0.026$, respectively), with no differences between BOO groups. This shows

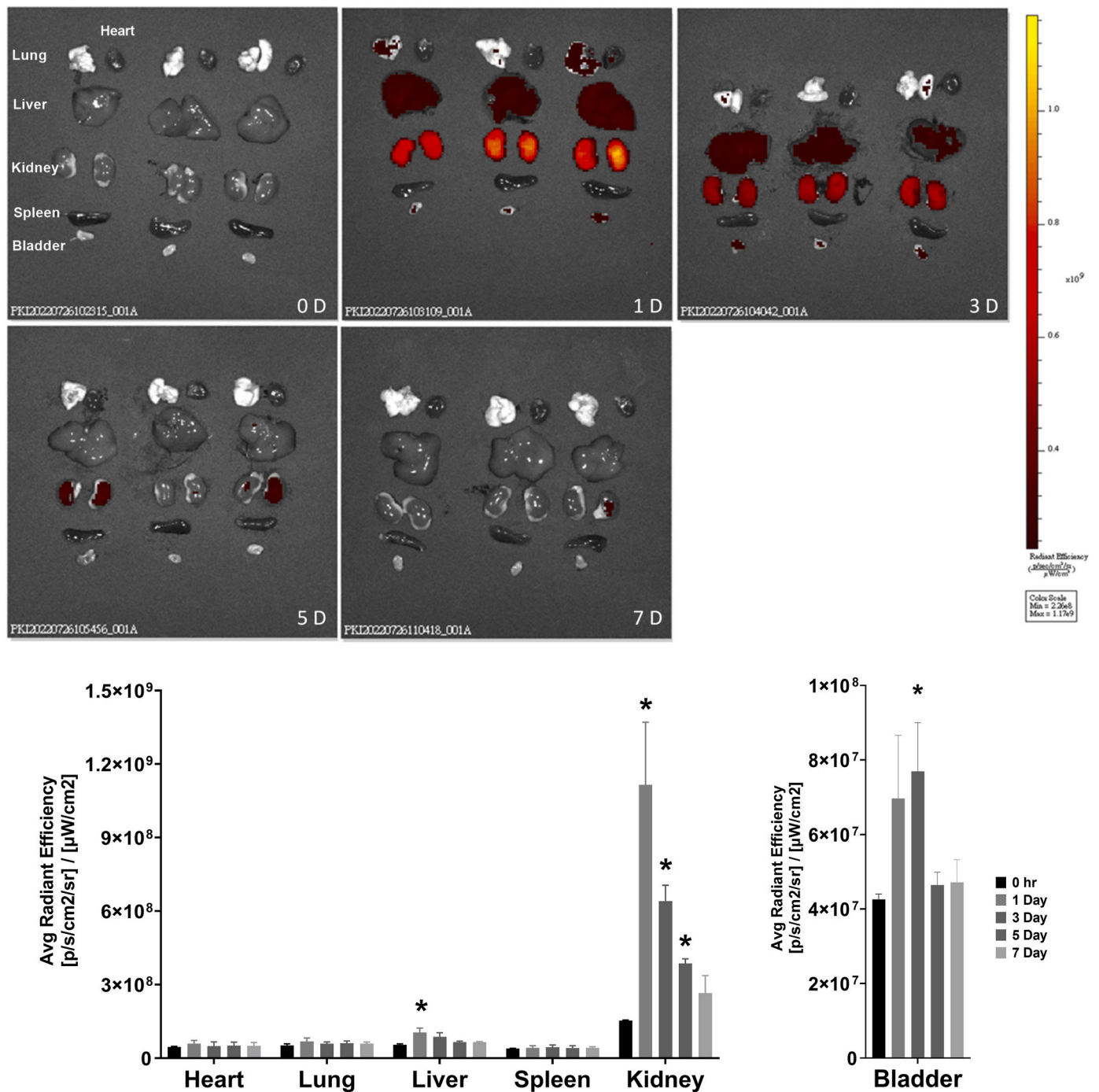


Fig. 5. Biodistribution assay of KCF18 and fluorescence signal intensity analysis, using rhodamine B-conjugated KCF18. A total of 15 adult male mice received a single subcutaneous injection of KCF-hydrogel (100 μL), containing 1.5 mg/kg KCF18 in a thermos-sensitive polyethylene glycol-based hydrogel (20 %, 5:5). Mice were sacrificed at baseline (0 h) and on Days 1, 3, 5, and 7 post-injection (n = 3 per time point). The heart, lungs, liver, kidneys, spleen, and bladder were collected for analysis. Biodistribution was assessed using an in vivo imaging system. Fluorescence signal intensity data from three mice per time point are presented as the mean ± standard deviation and compared to baseline (0 h). $p < 0.05$ vs. 0 h.

cytokine upregulation over time.

Fig. 7(B) shows fibrosis marker expression. TGF-β levels were higher in all BOO groups compared to Sham ($p = 0.015$, $p = 0.019$ and $p = 0.023$, respectively). There was no significant difference between BOO-14D and BOO-28D, but both were significantly higher than BOO-7D ($p = 0.024$ and $p = 0.023$, respectively). Collagen I followed the TGF-β pattern, while vimentin was elevated in all BOO groups ($p = 0.023$, $p = 0.003$ and $p = 0.014$, respectively), with no differences between BOO groups. α-SMA showed a similar trend, with no difference between BOO-

7D and Sham. This indicates fibrosis marker upregulation at all stages.

These data demonstrate that by Day 7 (early stage), BOO induces bladder dysfunction along with upregulation of cytokines (TNF-α, IL-1β, IL-6) and fibrosis markers (TGF-β, collagen I, vimentin). By Day 14 (mid-stage), bladder wall fibrosis and further dysfunction develop, with sustained elevation of cytokines and fibrosis markers. By Day 28 (late stage), bladder function continues to decline, cytokine levels remain elevated, and fibrosis markers stay high.

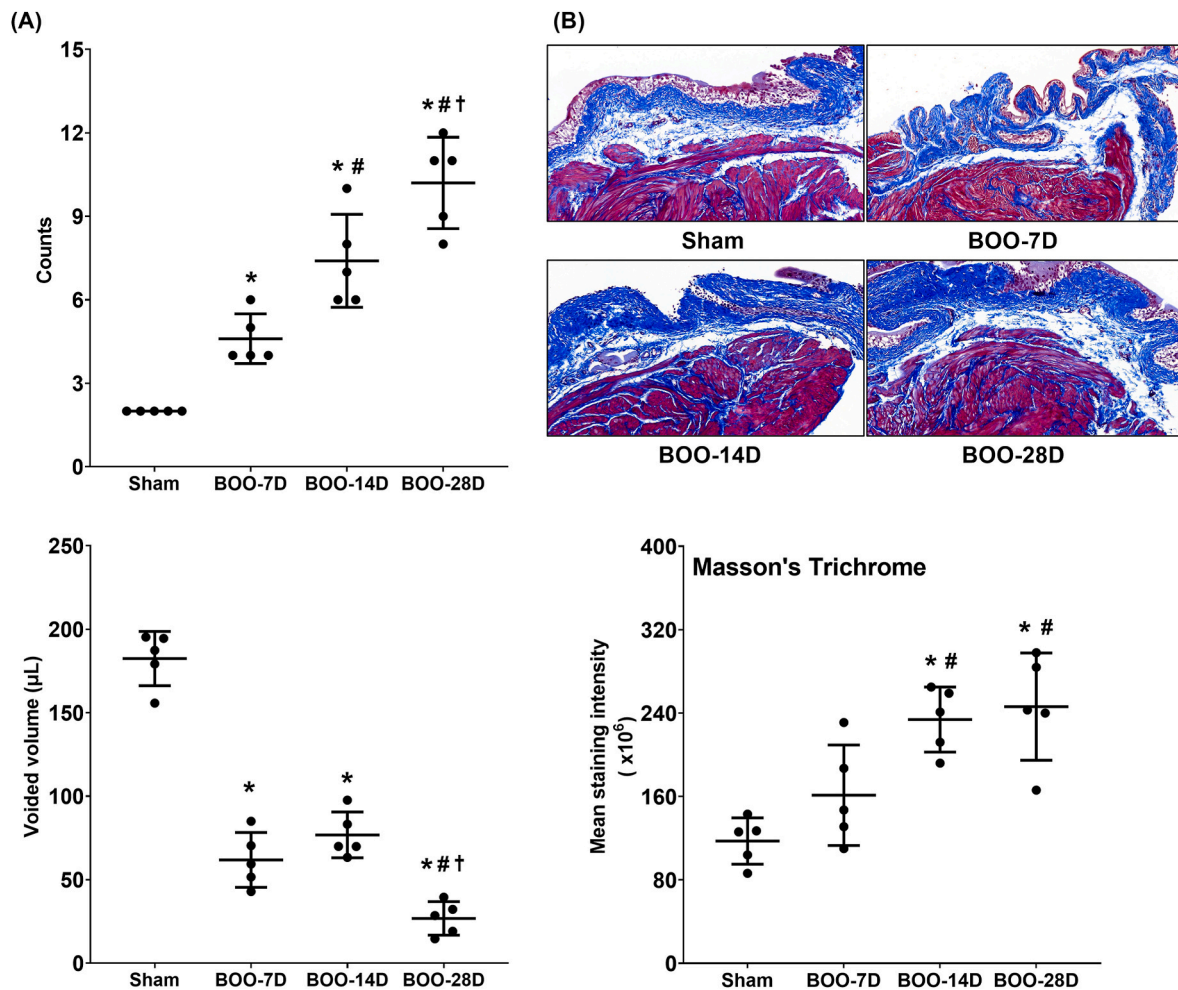


Fig. 6. Disease progression of bladder fibrosis induced by partial bladder outlet obstruction (BOO) in adult male mice. (A) Bladder function deterioration was assessed through voiding behavior analysis, measuring urine spot count (urinary frequency) and spot area (voiced volume). (B) Bladder fibrosis was evaluated using Masson's trichrome staining, with collagen deposition (blue staining) quantified. Bladder fibrosis was assessed on Days 7, 14, and 28 post-BOO to track disease progression. Data are presented as the mean $\pm$ standard deviation from five mice per time point. Sham: mice receiving a sham operation. BOO-7D: mice receiving BOO for 7 days. BOO-14D: mice receiving BOO for 14 days. BOO-28D: mice receiving BOO for 28 days. * $p < 0.05$ vs. Sham. # $p < 0.05$ BOO-14D or BOO-28D vs. BOO-7D. † $p < 0.05$ BOO-28D vs. BOO-14D. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3.3. Therapeutic effects of KCF-hydrogel, initiated in the early stage of bladder fibrosis, against BOO-induced bladder fibrosis in mice

3.3.1. KCF-hydrogel inhibits cytokine-receptor bindings in bladders of BOO mice

Fig. 8 shows cytokine-receptor binding levels in bladder tissues of mice following BOO surgery. Mice in the BOO-7D + KCF-H and BOO groups received weekly doses of KCF-hydrogel or normal saline for 7 consecutive weeks, starting on Day 7 post-BOO. Our analysis revealed that, on Day 56, fluorescence signals for TNF- α /TNFR1, IL-1 β /IL-1R, and IL-6/IL-6R binding were significantly lower in the BOO-7D + KCF-H group compared to the BOO group (all $p < 0.001$).

3.3.2. KCF-hydrogel inhibits bladder dysfunction and fibrosis in BOO mice

Fig. 9(A) shows voiding behavior analysis on Day 56 following BOO or sham surgery, with weekly KCF18-hydrogel therapy starting on Day 7. The BOO group had significantly more urine spots than Sham ($p < 0.001$), while the BOO-7D + KCF-H group showed significantly fewer spots compared to BOO ($p < 0.001$). Spot areas in the BOO group were smaller than Sham ($p = 0.003$), but the BOO-7D + KCF-H group had larger areas than BOO ($p = 0.015$). Notably, data in the Sham, Sham + H and Sham + KCF-H groups were comparable. Data in the BOO-7D + H and BOO groups were also comparable.

and BOO groups were also comparable.

Fig. 9(B) shows Masson's trichrome staining for collagen deposition. The BOO group had more intense blue staining than Sham ($p < 0.001$), while staining intensities in the Sham, Sham + H and Sham + KCF-H groups were similar. Notably, the BOO-7D + KCF-H group showed significantly less blue staining than the BOO group ($p < 0.001$), highlighting KCF18-hydrogel's effect in reducing fibrosis. Notably, data in the BOO-7D + H and BOO groups were also comparable.

3.3.3. KCF-hydrogel inhibits upregulation of cytokines in bladders of BOO mice

Fig. 10 presents cytokine levels in bladder tissues, measured by ELISA. TNF- α , IL-1 β , and IL-6 were significantly higher in the BOO group compared to Sham (all $p < 0.001$). In contrast, the BOO-7D + KCF-H group had significantly lower cytokine levels than BOO ($p < 0.001$ for TNF- α and IL-1 β , $p = 0.012$ for IL-6), demonstrating KCF18-hydrogel's ability to reduce cytokine upregulation. Notably, data in the Sham, Sham + H and Sham + KCF-H groups were comparable. Data in the BOO-7D + H and BOO groups were also comparable.

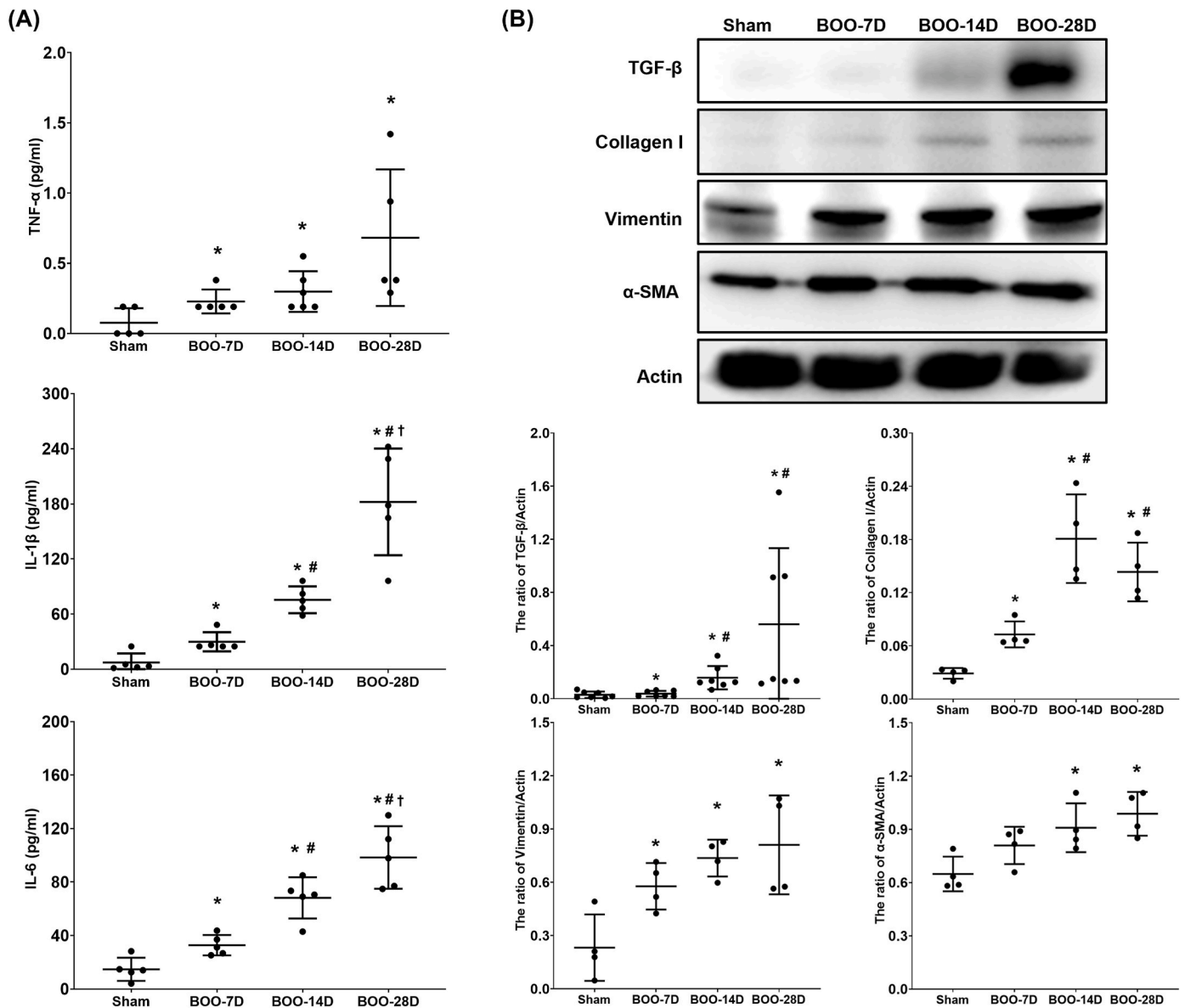


Fig. 7. Progression of cytokine and fibrosis marker expression in bladders of adult male mice with partial bladder outlet obstruction (BOO). (A) Cytokine concentrations, including tumor necrosis factor- α (TNF- α), interleukin-1 β (IL-1 β), and IL-6, were measured in bladder tissues using enzyme-linked immunosorbent assay (ELISA). (B) Representative immunoblot and densitometric analysis of fibrosis markers—transforming growth factor- β (TGF- β), collagen I, vimentin, and α -smooth muscle actin (α -SMA)—normalized to Actin as an internal control. Cytokine and fibrosis marker expression levels were assessed on Days 7, 14, and 28 post-BOO. Data are presented as mean $\pm$ standard deviation. ELISA data were obtained from five mice per time point, and immunoblotting data from four mice per time point. Sham: mice receiving sham surgery. BOO-7D: mice receiving BOO for 7 days. BOO-14D: mice receiving BOO for 14 days. BOO-28D: mice receiving BOO for 28 days. * p < 0.05 vs. Sham. # p < 0.05 BOO-14D or BOO-28D vs. BOO-7D. † p < 0.05 BOO-28D vs. BOO-14D.

3.3.4. KCF-hydrogel inhibits upregulation of fibrosis markers in bladders of BOO mice

Fig. 11 shows immunoblotting results for fibrosis markers. TGF- β , collagen I, and vimentin were significantly higher in the BOO group than Sham (all p < 0.001), but lower in the BOO-7D + KCF-H group compared to BOO (p = 0.040, p < 0.001 and p = 0.005, respectively). α -SMA expression followed a similar trend, except that the difference between the BOO and Sham groups was not statistically significant.

Immunohistochemistry (Fig. 12) confirmed these findings. Moreover, significantly higher α -SMA level in the BOO group compared to Sham (p < 0.001) was observed. Notably, data in the Sham, Sham + H and Sham + KCF-H groups were comparable. Data in the BOO-7D + H and BOO groups were also comparable.

These data collectively demonstrate that weekly KCF18-hydrogel administration for 7 weeks in the early stage of BOO significantly

mitigates bladder dysfunction, wall fibrosis, and upregulation of cytokines and fibrosis markers in mice.

3.4. Therapeutic effects of KCF-hydrogel, initiated in the mid- and late stage of bladder fibrosis, against bladder fibrosis in BOO mice

Fig. 13(A) shows voiding behavior analysis on Days 63 and 77 post-BOO surgery, with weekly KCF18-hydrogel therapy starting on Day 14 (mid-stage, BOO-14D + KCF-H) and Day 28 (late stage, BOO-28D + KCF-H), respectively. The BOO-14D + KCF-H group had significantly fewer spots and larger spot areas compared to the BOO group (both p < 0.001). In contrast, there were no significant differences in spot counts or areas between the BOO-28D + KCF-H and BOO groups, though the BOO-28D + KCF-H group had more spots and smaller areas compared to the BOO-14D + KCF-H group (p = 0.002 and p < 0.001, respectively).

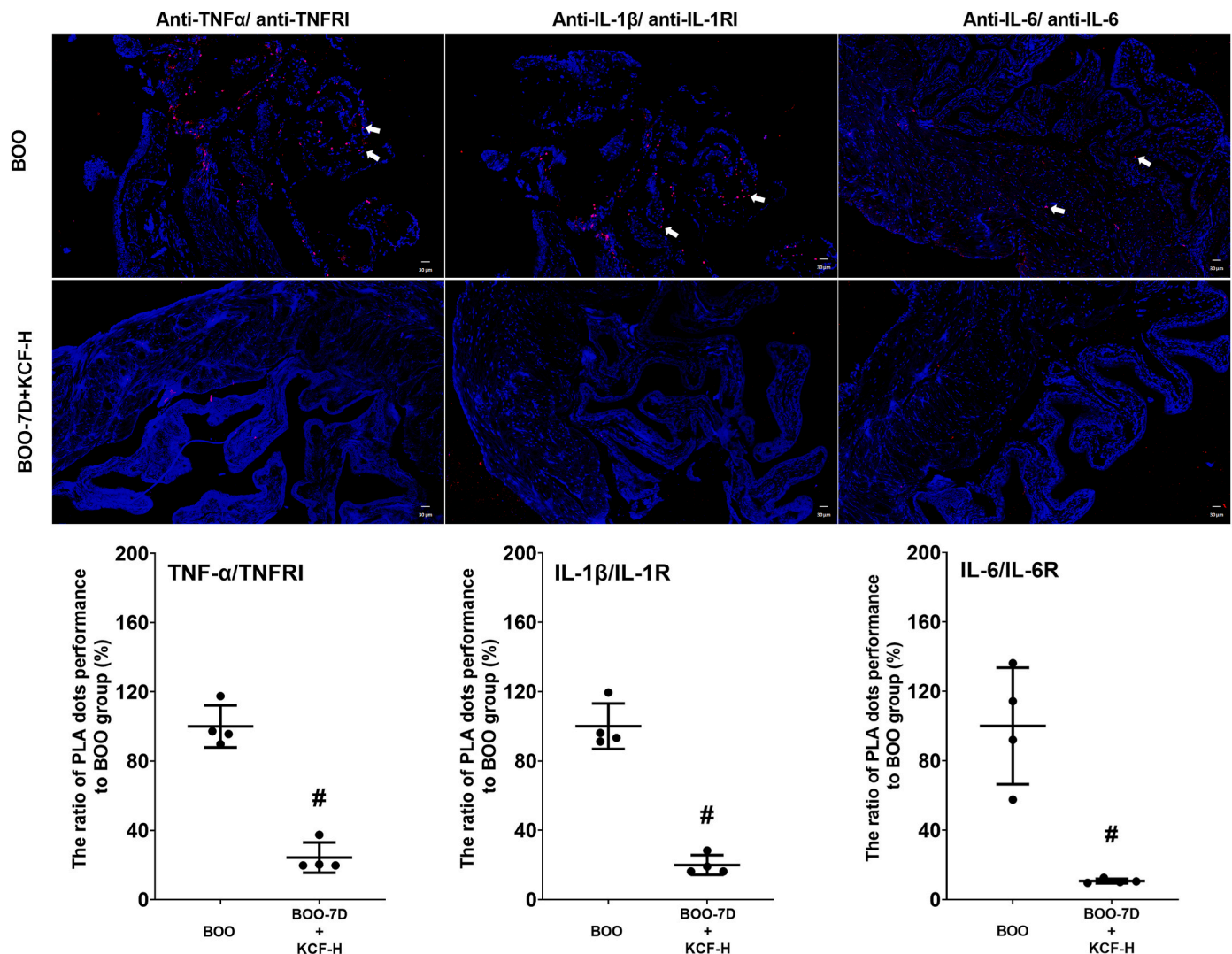


Fig. 8. Effects of KCF-hydrogel treatment, initiated in the early stage of bladder fibrosis, on inhibiting cytokine-receptor binding in bladders of adult male mice with partial bladder outlet obstruction (BOO). Mice ($n = 8$) were randomly assigned to the BOO or BOO-7D + KCF-H group ($n = 4$ per group). Cytokine-receptor binding, including tumor necrosis factor- α (TNF- α) with TNF receptor 1 (TNFR1), interleukin-1 β (IL-1 β) with IL-1 receptor (IL-1R), and IL-6 with IL-6 receptor (IL-6R), was assessed in bladder tissues using the proximity ligation assay (PLA). Representative PLA images and pink fluorescence signal intensities (white arrows) are shown. BOO: BOO mice receiving 100 μ L normal saline. BOO-7D + KCF-H: BOO mice receiving 100 μ L KCF-hydrogel, containing 1.5 mg/kg KCF18 in a thermos-sensitive polyethylene glycol-based hydrogel (20 %, 5:5). Weekly subcutaneous doses of normal saline and KCF-hydrogel were administered for seven weeks, beginning on Day 7 post-BOO and evaluated on Day 56. Data are presented as the mean $\pm$ standard deviation. # $p < 0.05$ BOO-7D + KCF-H vs. BOO. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Fig. 13(B) shows Masson's trichrome staining for collagen deposition. The BOO-14D + KCF-H group had significantly less blue staining than the BOO group ($p = 0.001$), whereas the BOO-28D + KCF-H and BOO groups showed no significant difference. Figs. 14 and 15 show similar patterns in cytokine expression analyzed by ELISA and fibrosis marker expression analyzed by immunohistochemistry.

Collectively, these data demonstrate that weekly KCF-hydrogel therapy starting in the mid-stage, but not the late stage, significantly mitigates BOO-induced bladder dysfunction, wall fibrosis, and upregulation of cytokines and fibrosis markers in mice.

4. Discussion

The data from this study demonstrate that the thermo-sensitive PEG-based hydrogels facilitate the controlled release of KCF18, a peptide decoy that inhibits TNF- α , IL-1 β , and IL-6 receptor binding [21]. Among the four formulations tested, the 20 % KCF-hydrogel (5:5) demonstrated the best KCF18 release profile, outperforming the 25 % (5:5), 25 % (4:6),

and 20 % (4:6) formulations. The study also shows that BOO induces significant bladder fibrosis and upregulation of inflammatory cytokines (TNF- α , IL-1 β , and IL-6), TGF- β (a master regulator of fibrosis), and markers of fibrotic processes, including collagen I (for ECM remodeling), α -SMA (for FMT), and vimentin (for EMT) [18,19]. These effects become apparent by Day 7 (early stage) and worsen by Days 14 and 28 (mid- and late stages). Furthermore, the data highlight that continuous inhibition of TNF- α , IL-1 β and IL-6 receptor binding, mediated by the controlled release of KCF18 from the thermo-sensitive PEG-based hydrogel (20 %, 5:5), can mitigate bladder fibrosis and suppress the upregulation of TGF- β and fibrosis markers induced by BOO in mice. These findings support our hypothesis and emphasize the potential of targeting the upregulation of inflammatory cytokines as a therapeutic strategy for BOO-induced bladder fibrosis. Notably, therapeutic benefits are observed when treatment is initiated during the early or mid-stages of BOO, but not during the late stage. These results further underline the importance of timely intervention in this context. Given the current lack of effective therapies to inhibit BOO-induced bladder fibrosis, the data

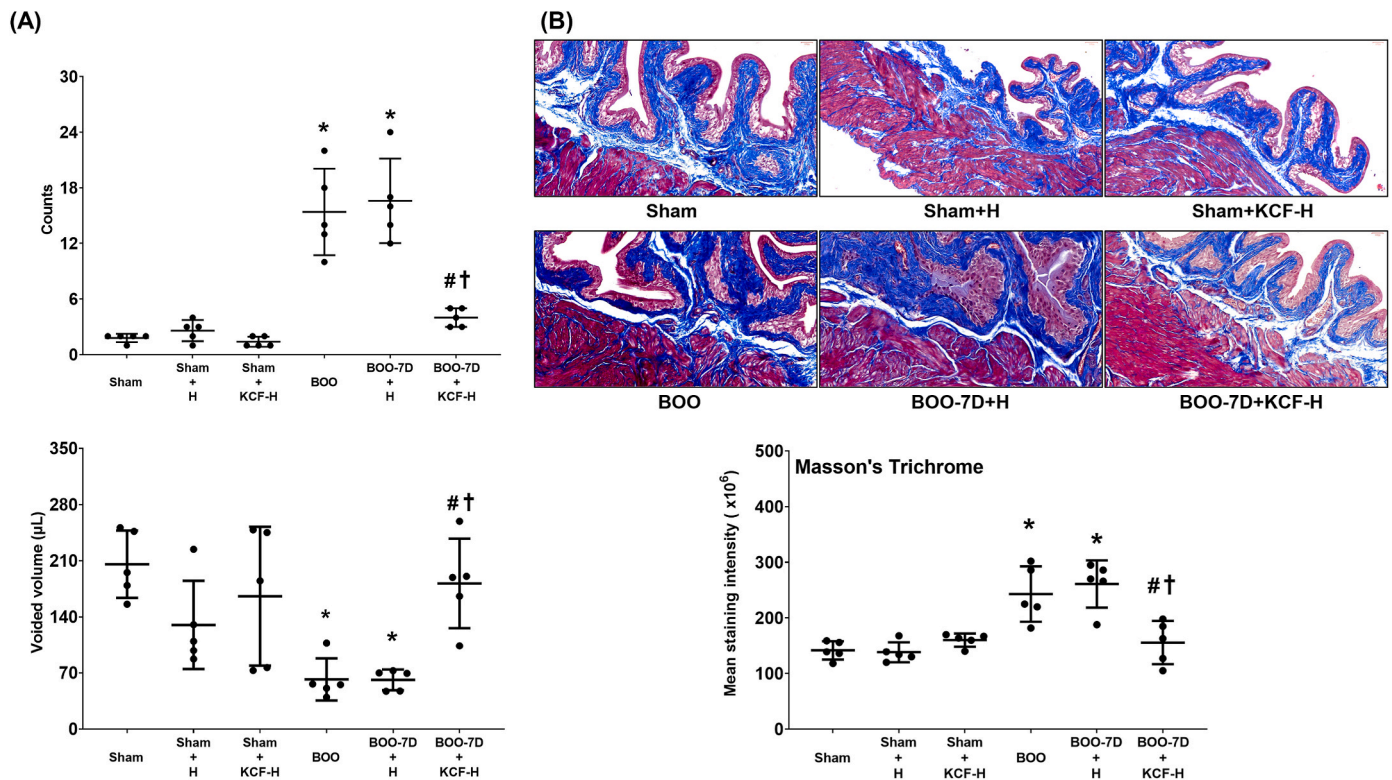


Fig. 9. Effects of KCF-hydrogel treatment, initiated in the early stage of bladder fibrosis, on bladder dysfunction and fibrosis in adult male mice with partial bladder outlet obstruction (BOO). (A) Bladder function deterioration was assessed through voiding behavior analysis, measuring urine spot count (urinary frequency) and spot area (voided volume). (B) Bladder fibrosis was evaluated using Masson's trichrome staining, with collagen deposition (blue staining) quantified. Weekly subcutaneous injections were administered for seven weeks, beginning on Day 7 post-sham or BOO surgery, with evaluations conducted on Day 56. Data are presented as mean $\pm$ standard deviation from five mice per group. Sham: sham-operated mice receiving 100 μ L normal saline. Sham + H: sham-operated mice receiving 100 μ L thermos-sensitive polyethylene glycol [PEG]-based hydrogel (20 %, 5:5). Sham + KCF-H: sham-operated mice receiving 100 μ L KCF-hydrogel, containing 1.5 mg/kg KCF18 in a thermosensitive PEG-based hydrogel (20 %, 5:5). BOO: BOO mice receiving 100 μ L normal saline. BOO-7D + H: BOO mice receiving 100 μ L hydrogel. BOO-7D + KCF-H: BOO mice receiving 100 μ L KCF-hydrogel. * p < 0.05 vs. Sham. # p < 0.05 vs. BOO. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

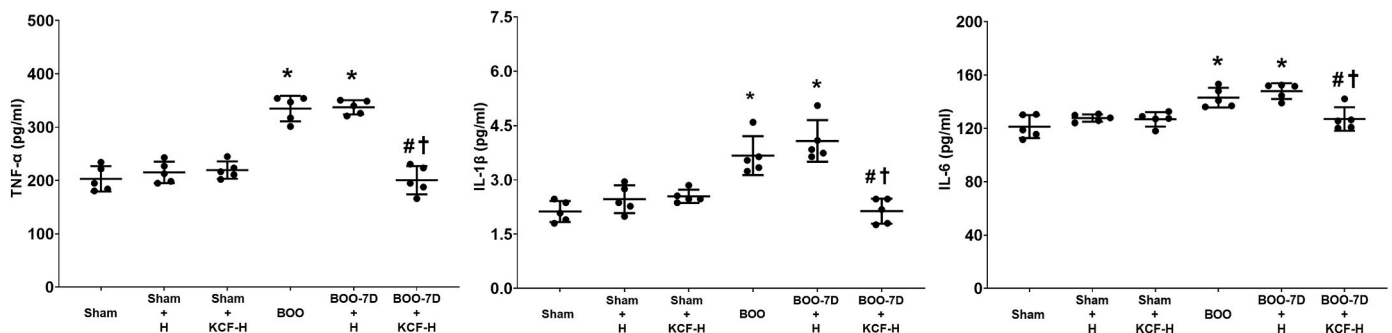


Fig. 10. Effects of KCF-hydrogel treatment, initiated in the early stage of bladder fibrosis, on inflammatory cytokine levels in bladders of adult male mice with partial bladder outlet obstruction (BOO). Tumor necrosis factor- α (TNF- α), interleukin-1 β (IL-1 β), and IL-6 levels were measured using enzyme-linked immunosorbent assay (ELISA). Weekly subcutaneous injections were administered for seven weeks, beginning on Day 7 post-sham or BOO surgery, with evaluations conducted on Day 56. Data are presented as mean $\pm$ standard deviation from five mice per group. Sham: sham-operated mice receiving 100 μ L normal saline. Sham + H: sham-operated mice receiving 100 μ L thermos-sensitive polyethylene glycol (PEG)-based hydrogel (20 %, 5:5). Sham + KCF-H: sham-operated mice receiving 100 μ L KCF-hydrogel, containing 1.5 mg/kg KCF18 in a thermosensitive PEG-based hydrogel (20 %, 5:5). BOO: BOO mice receiving 100 μ L normal saline. BOO-7D + H: BOO mice receiving 100 μ L hydrogel. BOO-7D + KCF-H: BOO mice receiving 100 μ L KCF-hydrogel. * p < 0.05 vs. Sham. # p < 0.05 vs. BOO.

from this study have significant implications, highlighting the need for further investigation in this area.

The central mechanism appears to be the inflammatory response triggered by BOO. This inflammatory process induces upregulation of TGF- β , which in turn triggers the fibrotic processes and ultimately resulting in fibrosis [18,19]. KCF18 plays a crucial role in inhibiting this process. By inhibiting the binding of cytokines (TNF- α , IL-1 β , and IL-6)

to their respective receptors (TNFR1, IL-1R, and IL-6R) [21], KCF18 effectively interrupts this inflammatory cascade, preventing subsequent fibrotic changes in the bladder. As bladder fibrosis is alleviated, intravesical pressure during urination decreases, and the inflammatory response is suppressed [13,14]. Our data show that KCF-hydrogel-treated BOO male mice had lower levels of TNF- α , IL-1 β , and IL-6 in bladders compared to untreated BOO mice, which correlated

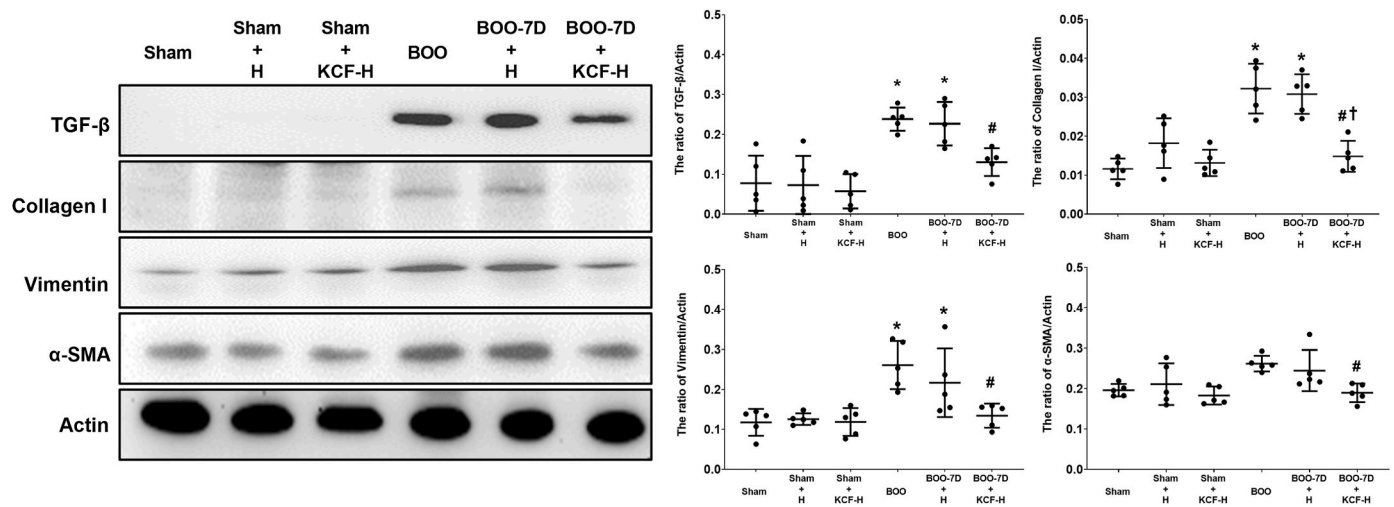


Fig. 11. Effects of KCF-hydrogel treatment, initiated in the early stage of bladder fibrosis, on the expression of fibrosis-associated markers in bladders of adult male mice with partial bladder outlet obstruction (BOO), assessed via immunoblotting. Representative gel images and densitometry data showing the expression of transforming growth factor-β (TGF-β), collagen I, vimentin, and α-smooth muscle actin (α-SMA) are presented. Weekly subcutaneous injections were administered for seven weeks, beginning on Day 7 post-sham or BOO surgery, with evaluations conducted on Day 56. Data are presented as mean ± standard deviation from four mice per group. Sham: sham-operated mice receiving 100 μL normal saline. Sham + H: sham-operated mice receiving 100 μL thermos-sensitive polyethylene glycol (PEG)-based hydrogel (20 %, 5:5). Sham + KCF-H: sham-operated mice receiving 100 μL KCF-hydrogel, containing 1.5 mg/kg KCF18 in a thermosensitive PEG-based hydrogel (20 %, 5:5). BOO: BOO mice receiving 100 μL normal saline. BOO-7D + H: BOO mice receiving 100 μL hydrogel. BOO-7D + KCF-H: BOO mice receiving 100 μL KCF-hydrogel. *p < 0.05 vs. Sham. #p < 0.05 vs. BOO.

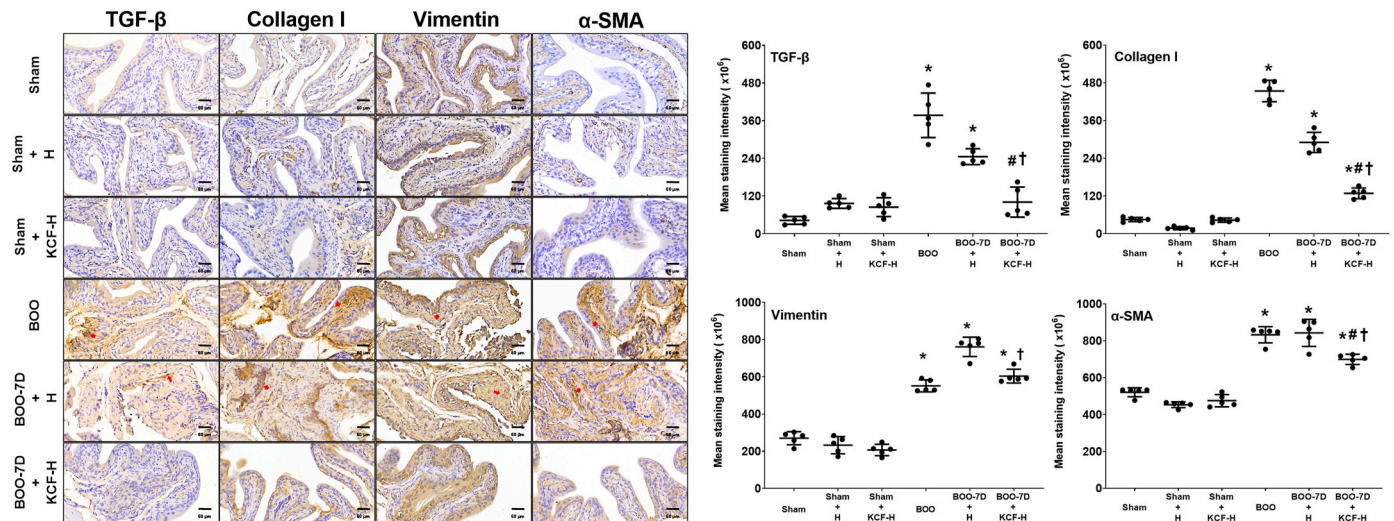


Fig. 12. Effects of KCF-hydrogel treatment, initiated in the early stage of bladder fibrosis, on the expression of fibrosis-associated markers in bladders of adult male mice with partial bladder outlet obstruction (BOO), assessed via immunohistochemistry. Representative photomicrographs and signal densities of brown staining (marked by red arrows) showing the expression of transforming growth factor-β (TGF-β), collagen I, vimentin, and α-smooth muscle actin (α-SMA) are presented. Weekly subcutaneous injections were administered for seven weeks, starting on Day 7 post-sham or BOO surgery, with evaluations conducted on Day 56. Data are presented as mean ± standard deviation from five mice per group. Sham: sham-operated mice receiving 100 μL normal saline. Sham + H: sham-operated mice receiving 100 μL thermos-sensitive polyethylene glycol (PEG)-based hydrogel (20 %, 5:5). Sham + KCF-H: sham-operated mice receiving 100 μL KCF-hydrogel, containing 1.5 mg/kg KCF18 in a thermosensitive PEG-based hydrogel (20 %, 5:5). BOO: BOO mice receiving 100 μL normal saline. BOO-7D + H: BOO mice receiving 100 μL hydrogel. BOO-7D + KCF-H: BOO mice receiving 100 μL KCF-hydrogel. *p < 0.05 vs. Sham. #p < 0.05 vs. BOO. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

with reduced TGF-β and fibrosis marker expression in the bladders. These results highlight the importance of interrupting inflammatory mechanisms to mitigate the progression of bladder fibrosis.

Peptides, essential components of proteins formed by amino acids connected via peptide bonds, present a sustainable alternative to synthetic drugs owing to their remarkable target specificity, potency, and biodegradability [24]. Their versatile applications encompass the treatment of various diseases, such as inflammatory diseases, cardiovascular diseases and breast cancer [20–22,38], along with their

potential as antibiotics due to their inherent antimicrobial properties [24]. Furthermore, breakthroughs like KCF18 peptide decoy [21] showcases its efficacy in targeting specific cytokine-receptor interactions and adaptable customization to suit diverse biological needs. Nonetheless, peptides face a significant challenge—they are prone to degradation by proteolytic enzymes, resulting in a very short in vivo half-life [24], which currently hampers their widespread clinical utilization.

As mentioned earlier, this study utilized a thermos-sensitive PEG-

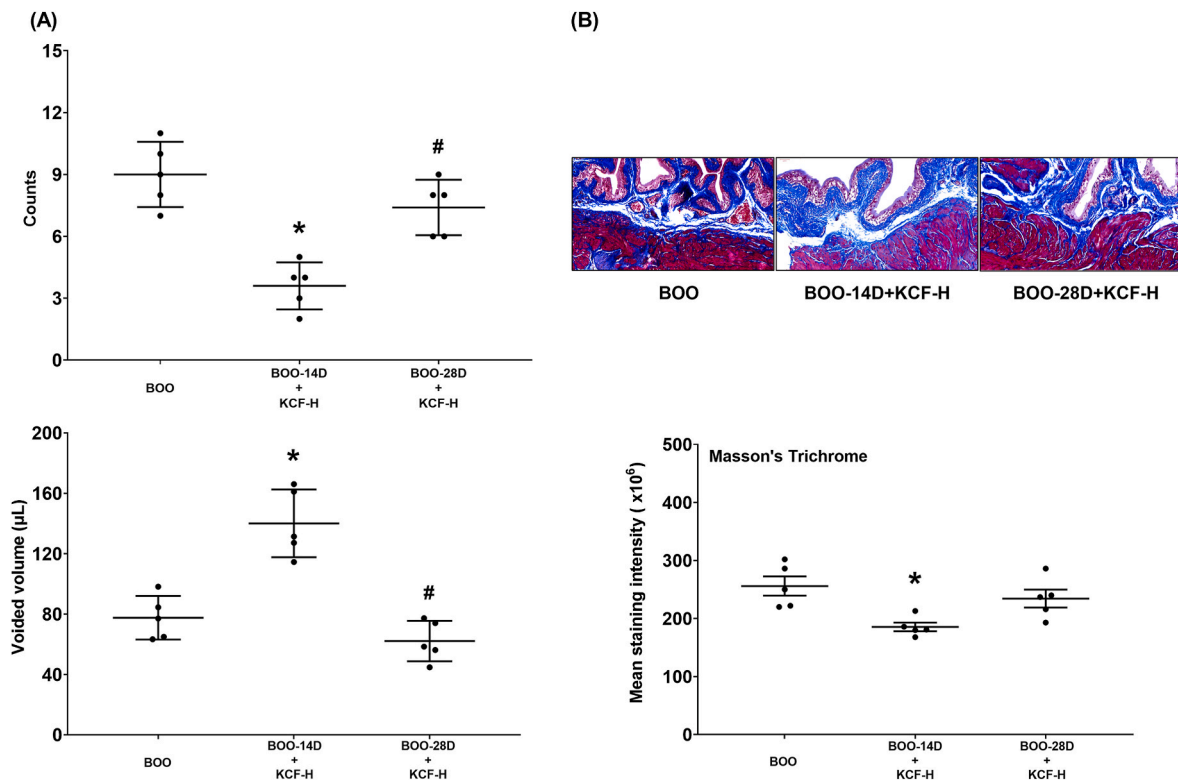


Fig. 13. Effects of KCF-hydrogel treatment, initiated in the mid- and late stages of bladder fibrosis, on alleviating bladder dysfunction and fibrosis in adult male mice with partial bladder outlet obstruction (BOO). (A) Bladder function deterioration was assessed by voiding behavior analysis, including urine spot count (urinary frequency) and spot area (voided volume). (B) Bladder fibrosis was evaluated using Masson's trichrome staining to visualize collagen deposition (blue staining), with signal intensities quantified. Weekly subcutaneous injections were administered for seven weeks, starting on Days 14 and 28 post-BOO surgery, with evaluations conducted on Days 63 and 77 respectively. Data are presented as mean $\pm$ standard deviation from five mice per group. BOO: BOO mice receiving 100 μ L normal saline. BOO-14D + KCF-H: BOO mice receiving weekly dose of 100 μ L KCF-hydrogel (20 %, 5:5) for seven weeks, containing 1.5 mg/kg KCF18 in a thermos-sensitive polyethylene glycol-based hydrogel (20 %, 5:5), starting on Day 14 post-BOO. BOO-28D + KCF-H: BOO mice receiving weekly dose of 100 μ L KCF-hydrogel, starting on Day 28 post-BOO. * p < 0.05 vs. BOO. # p < 0.05 BOO-28D + KCF-H vs. BOO-14D + KCF-H. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

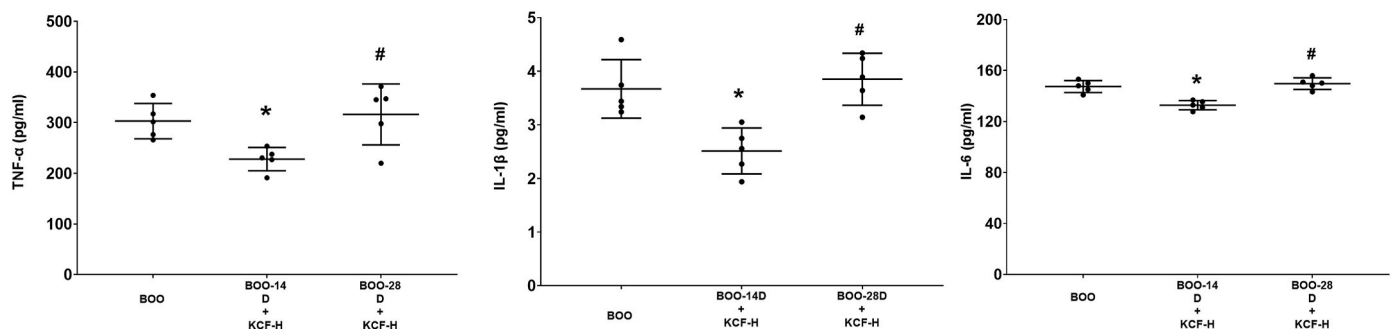


Fig. 14. Effects of KCF-hydrogel treatment, initiated in the mid and late stages of bladder fibrosis, on inflammatory cytokines (tumor necrosis factor- α [TNF- α], interleukin-1 β [IL-1 β], and IL-6) in bladders of adult male mice with partial bladder outlet obstruction (BOO), measured using enzyme-linked immunosorbent assay (ELISA). Weekly subcutaneous injections were administered for seven weeks, starting on Days 14 and 28 post-BOO surgery, with evaluations on Days 63 and 77, respectively. Data are presented as mean $\pm$ standard deviation from five mice per group. BOO: BOO mice receiving 100 μ L normal saline. BOO-14D + KCF-H: BOO mice receiving weekly dose of 100 μ L KCF-hydrogel (20 %, 5:5) for seven weeks, containing 1.5 mg/kg KCF18 in a thermos-sensitive polyethylene glycol-based hydrogel (20 %, 5:5), starting on Day 14 post-BOO. BOO-28D + KCF-H: BOO mice receiving weekly dose of 100 μ L KCF-hydrogel, starting on Day 28 post-BOO. * p < 0.05 vs. BOO. # p < 0.05 BOO-28D + KCF-H vs. BOO-14D + KCF-H.

based hydrogel to achieve a controlled release of the peptide decoy KCF18. Hydrogels, renowned for their water-absorbing capacity due to their intricate cross-linked structure, are extensively utilized in controlled drug delivery [39]. Their porous design enables the encapsulation of diverse therapeutic agents, facilitating tailored and prolonged release for conditions such as bladder fibrosis and malignancies [25,27,39]. Their responsiveness to external stimuli, including

temperature, makes them ideal for targeted drug delivery [26,27,33,40]. Injectable thermos-sensitive hydrogels, with their temperature-induced phase transition, hold promise in disease treatments [26,27,33,40]. Integration of KCF18 with the thermos-sensitive PEG-based hydrogel in this study emphasizes its sustained release properties. Our preliminary data showed that weekly injections of KCF-hydrogel for four weeks significantly mitigated BOO-induced

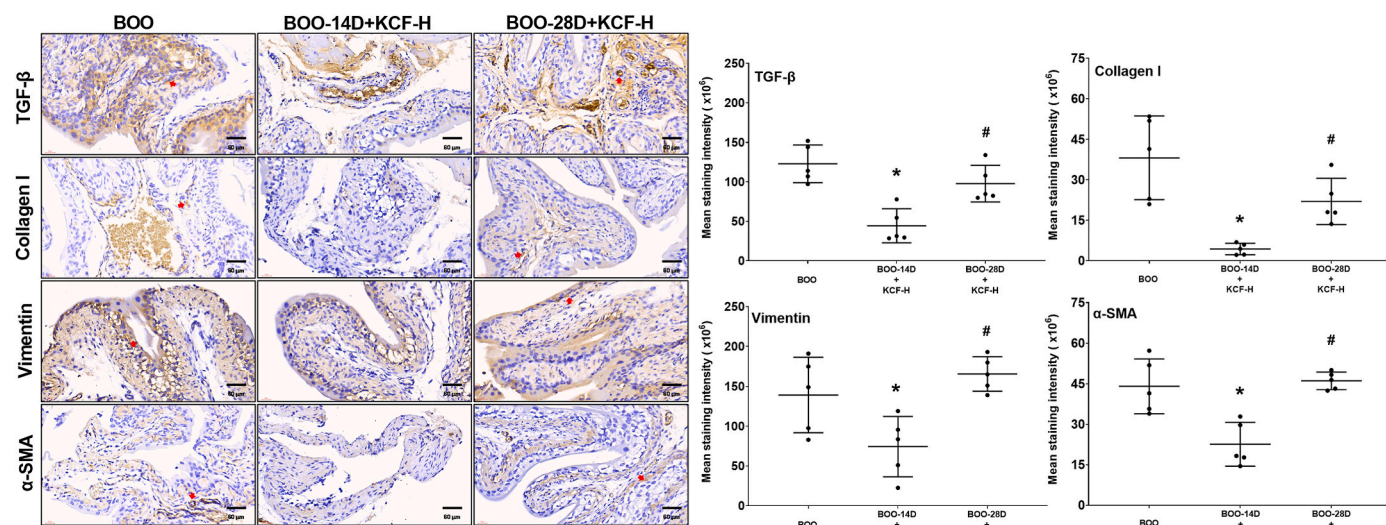


Fig. 15. Effects of KCF-hydrogel treatment, initiated in the mid and late phases of bladder fibrosis, on the upregulation of transforming growth factor- β (TGF- β) and fibrosis-associated markers (collagen I, vimentin, and α -smooth muscle actin [α -SMA]) in bladders of adult male mice with partial bladder outlet obstruction (BOO), assessed via immunohistochemistry. Representative photomicrographs and signal densities of brown staining (marked by red arrows) are shown. Weekly subcutaneous injections were administered for seven weeks, starting on Days 14 and 28 post-BOO surgery, with evaluations conducted on Days 63 and 77, respectively. Data are presented as mean $\pm$ standard deviation from five mice per group. BOO: BOO mice receiving 100 μ L normal saline. BOO-14D + KCF-H: BOO mice receiving weekly dose of 100 μ L KCF-hydrogel (20 %, 5:5) for seven weeks, containing 1.5 mg/kg KCF18 in a thermos-sensitive polyethylene glycol-based hydrogel (20 %, 5:5), starting on Day 14 post-BOO. BOO-28D + KCF-H: BOO mice receiving weekly dose of 100 μ L KCF-hydrogel, starting on Day 28 post-BOO. * p < 0.05 vs. BOO. # p < 0.05 BOO-28D + KCF-H vs. BOO-14D + KCF-H. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

bladder dysfunction and fibrosis, whereas weekly injections of KCF18 without hydrogel conjugation did not yield significant therapeutic effects (please refer to [Supplemental Fig. 1B](#)). These findings confirm the therapeutic benefit of the hydrogel-based delivery strategy. Moreover, this approach may also address the challenge of frequent drug administration, particularly relevant in cases of suboptimal patient adherence and thereby expanding therapeutic possibilities.

As highlighted above, timely intervention is critical in managing BOO-induced bladder fibrosis. Our study demonstrates that disease progression significantly impacts the therapeutic efficacy of continuous cytokine inhibition, which works by blocking cytokine-receptor bindings. While cytokine inhibition mitigates bladder dysfunction and fibrosis in the early and mid-stages of BOO-induced bladder fibrosis, its effects are less pronounced in the late stage. One possible explanation for this is that, once TGF- β expression and bladder fibrosis are firmly established in the late stage, a vicious cycle of heightened intravesical pressure during urination and continuous upregulation of cytokines due to bladder fibrosis is set in motion. As a result, inhibiting cytokines from binding to their receptors may no longer effectively reduce inflammatory response as well as decrease TGF- β levels and modulate its activity in the late stage of BOO-induced bladder fibrosis. This hypothesis is supported by our data from the late stage of BOO-induced bladder fibrosis, where KCF-hydrogel treatment showed limited effects on modulating cytokine upregulation, upregulation of fibrosis markers and bladder fibrosis. Based on these findings, we conjecture that therapies targeting TGF- β directly—either by inhibiting its expression or blocking its interaction with the TGF- β receptor—may provide greater therapeutic benefits in the later stages of BOO-induced bladder fibrosis. Currently, our laboratory is conducting follow-up studies to explore the potential of novel peptide decoys that can block TGF- β receptor binding in this context.

Our study has several inherent limitations. Firstly, the bio-distribution of KCF18 in bladder tissues post-injection was suboptimal, suggesting that adjustments in dosing or administration techniques are needed to improve local delivery to the bladder. Moreover, a discrepancy between the fluorescence biodistribution and pharmacokinetic

profiles was observed, as KCF18 accumulation in the bladder peaked on Day 3, while plasma concentrations peaked between 24 and 48 h (Days 1–2). Although the underlying mechanisms remain unclear, this discrepancy may reflect differences in biological compartments and detection methods: plasma pharmacokinetics represent systemic drug levels, whereas IVIS imaging reflects local tissue accumulation. Following subcutaneous hydrogel administration, the drug is gradually released into circulation and subsequently distributed to tissues, making it plausible for plasma levels to peak earlier than tissue signals observed by IVIS. Further studies are needed to clarify this mechanism. Secondly, our preliminary data demonstrated that the KCF-hydrogel containing KCF18 at 1.5 mg/kg, but not at 0.6 mg/kg (please refer to [Supplemental Fig. 1C](#)), significantly attenuated BOO-induced bladder dysfunction and fibrosis. Furthermore, serum biochemistry analyses revealed no significant liver or kidney toxicity in mice treated with the KCF-hydrogel containing KCF18 1.5 mg/kg, supporting its *in vivo* biocompatibility (please refer to [Supplemental Fig. 1D](#)). Nevertheless, it remains to be determined whether doses higher than 1.5 mg/kg could confer additional therapeutic benefits in this context, and potential toxicity at higher doses also remains unstudied. Thirdly, among the four KCF-hydrogel formulations tested, the 20 % KCF-hydrogel (5:5) demonstrated the most favorable *in vitro* release efficacy. Based on this comparative release profile, we selected this formulation for further studies to ensure more efficient drug delivery within the critical early treatment window. While we acknowledge that longer release durations may be desirable in certain contexts, the observed performance of this formulation provided a practical balance between release efficacy and experimental feasibility. At this stage, our data did not indicate a definitive “incomplete release” issue with the other formulations; rather, the selection was guided by relative release performance. Nonetheless, the underlying mechanisms contributing to the superior release characteristics of the 20 % KCF-hydrogel (5:5) remain unclear. It is also uncertain whether alternative hydrogel formulations—such as those with modified polymer compositions or concentrations—could further enhance controlled release, highlighting the need for additional investigation.

Fourthly, previous studies have demonstrated the efficacy of anti-inflammatory therapies (e.g., metformin, mesenchymal stem cells, etc.) [41,42] in treating bladder fibrosis. In a BOO mouse model, metformin administered daily for 9 weeks restored bladder voiding efficiency to approximately 70 % of normal levels [41]. In comparison, our KCF-hydrogel therapy, administered weekly for 7 weeks, restored voiding efficiency to 86 %. In terms of anti-fibrotic effects, human adipose-derived stem cell therapy administered for 4 weeks reduced TGF- β expression to approximately 65 % [42], whereas our KCF-hydrogel therapy reduced TGF- β levels to around 33 % after 7 weeks of treatment. Although no direct head-to-head comparisons were conducted, these findings suggest that KCF-hydrogel therapy demonstrates robust therapeutic efficacy with a potentially more convenient dosing schedule and translational feasibility. Furthermore, monoclonal antibodies [43–45] or fusion proteins such as TAL-6 [46] targeting TNF- α , IL-1 β , and IL-6 represent another approach to cytokine inhibition. However, this study did not directly compare KCF18 with these treatments, raising the question of whether KCF-hydrogel may outperform them. Nevertheless, KCF18 offers the advantage of simultaneous inhibition of all three cytokines, potentially simplifying treatment and reducing costs. The data from this study thus lay the foundation for future research and clinical applications, offering hope for patients and underscoring the need for continued investigations. Lastly, bladder fibrosis can also occur in females, such as in cases of interstitial cystitis [47]. Whether KCF-hydrogel therapy can provide similar therapeutic benefits in this context remains to be explored.

In conclusion, our study demonstrates that partial bladder outlet obstruction drives bladder inflammation and fibrosis. Timely intervention with sustained cytokine inhibition in the early and mid-disease stages, enabled by thermo-sensitive PEG-based hydrogel-mediated controlled KCF18 peptide decoy release, effectively mitigates inflammation and reduces bladder fibrosis in male mice.

CRedit authorship contribution statement

Syuan-Hao Syu: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Chao-Yuan Chang:** Writing – review & editing, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Pu-Sheng Wei:** Conceptualization, Data curation, Formal analysis, Writing – review & editing. **Van Long Le:** Writing – review & editing, Formal analysis, Data curation, Conceptualization. **Yu-Ching Wen:** Writing – review & editing, Formal analysis, Data curation, Conceptualization. **Hao-Jen Hsu:** Writing – review & editing, Resources, Formal analysis, Conceptualization. **Hung-Jen Shih:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Chun-Jen Huang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of generative AI in scientific writing

During the preparation of this manuscript, the authors used ChatGPT (version 3.5) to assist with language refinement and improve the clarity and readability of the text. All content generated with the help of this tool was carefully reviewed and revised by the authors, who take full responsibility for the accuracy and integrity of the final version.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jddst.2025.107533>.

Data availability

Data will be made available on request.

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